

Camera Motion Control Project

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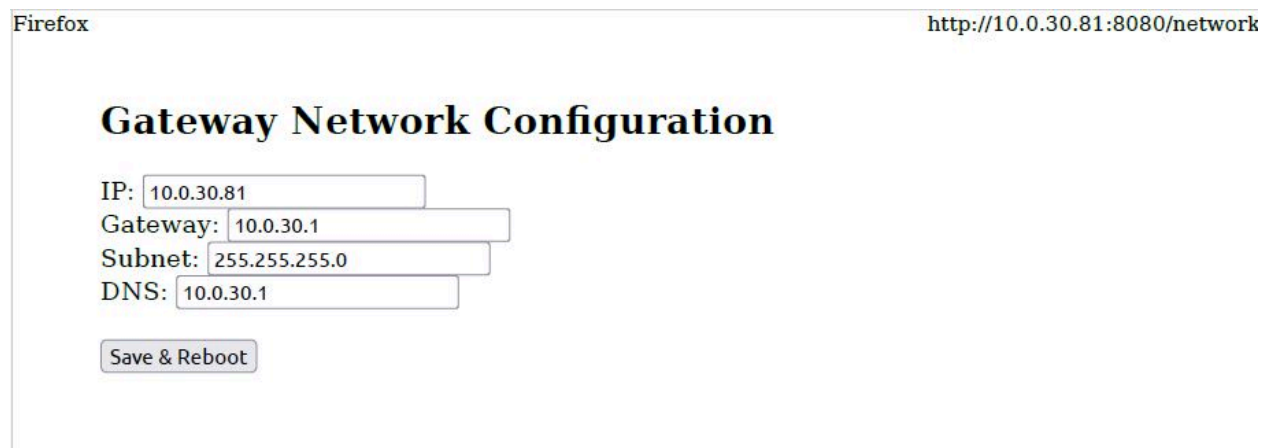
Updating the device IP

The device IP can be found displayed on a small OLED display attached to each device. If you need to change the IP you can do so from a browser window using the following command:

`http://10.0.30.81:8080/network`

The First part of this address is the current IP address of the device which you should see displayed on the small OLED display attached to the device mainboard by default this should be **10.0.30.81** however if this has been changed you should replace the Ip in the http address above to match the IP shown on the device display. Currently the system assumes you are running the config from a BST system with a gateway address of **10.0.30.1** if this is not the case you will not be able to reach the device as they may not be on the same network. This will be addressed on future updates.

Assuming your network can see the device you should see the following menu appear on your browser page:



The screenshot shows a web browser window with the title 'Firefox' and the address bar displaying 'http://10.0.30.81:8080/network'. The main content area has the heading 'Gateway Network Configuration'. Below the heading are four input fields for network configuration: 'IP:' with the value '10.0.30.81', 'Gateway:' with the value '10.0.30.1', 'Subnet:' with the value '255.255.255.0', and 'DNS:' with the value '10.0.30.1'. At the bottom of the form is a button labeled 'Save & Reboot'.

. Be careful to use an IP you can reach from your network to avoid **BRICKING** the device. There is some error checking but this needs to be made more robust going forward.

Once you have set a new IP address you should see the new address on the device OLED display

Slider General Mechanical test setup

The test slider used for developing and testing the system software had the following parameters

- 1) A nema17 stepper motor 1.8deg/step with a 20T GT2 pulley and timing belt
- 2) A simple homing switch at the motor end of the slider.
- 3) At startup the slider moves to find home with the carriage moving right to left towards the homing switch. The Motor is mounted at the same end of the slider as the motor. When the slider moves to find home the motor turns in a clockwise direction when looking from the front of the motor.
- 4) The carriage is capable of moving approx 345mm
- 5) The joystick should move the carriage in the correct corresponding direction

If your design differs from this in any way you will have to use the configuration menu in order to adjust the key parameters to match your particular configuration.

Slider Configuration Menu

In the absence of the final PTZrobot device the software for the slider has been tested and configured for the mechanics of the Digital Bird slider. PTZrobot will have their own requirements depending on the configuration of the final product. To this end I have provided a slider configuration menu which can be accessed from a browser window with the command:

http://10.0.30.81:8080/slider_config

The First part of this address is the current IP address of the slider which you should see displayed on the small OLED display attached to the device mainboard by default this should be **10.0.30.81** however if this has been changed you should replace the Ip in the http address above to match the IP shown on the device display. Currently the system assumes you are running the config from a BST system with a gateway address of **10.0.30.1** if this is not the case you will not be able to reach the device as they may not be on the same network. This will be addressed on future updates.

Assuming your network can see the device you should see the following menu appear on your browser page:

Firefox http://10.0.30.81:8080/slider_config

Slider Configuration

Mechanical Setup

Steps per rev:	200.00
Microsteps:	64
mm per rev:	34.90
Steps per mm:	366.76

Motion Settings

Max speed (mm/s):	25.00
Acceleration (mm/s ²):	200.00
Joystick max vel (mm/s):	200.00
Joystick accel (mm/s ²):	200.00
Invert direction:	☐

Travel Limits

Logical home offset (mm):	10.00
Soft max mm (max travel):	345.00

This gives you access to all the configurable settings for the slider firmware and the table below outlines their purpose.

Mechanical Setup	
steps_per_rev = 200	1,8deg stepper=200 0.9deg stepper=400
microsteps = 64	Fixed at 64 for now
mm_per_rev = 34.9	pulley/belt dependent (How far the carriage moves per one full revolution of the motor) Maths will get you close but will require fine tuning against what you actually measure on your device.
float steps_per_mm=321.3	Is the conversion factor between real-world millimetres and the internal floating-point "logical position" provided by the encoder. logical_mm=encoder_counts×float_per_mm Think of it as a scale factor for the encoder checked against real world moves.
Motion Settings	
maxSpeed_mm_s = 25	Limits the Maximum speed of the slider to sensible values
accel_mm_s2 = 600	Limits the Max acceleration of the slider to sensible values
bst_max_vel_mm_s = 200	Defines the top speed (in mm/s) that BST is allowed to command during joystick or live-drive control. It acts as a safety and feel limit so BST can't accidentally request speeds that are too fast for the mechanics.
bst_accel_mm_s2 = 200	The acceleration limit used specifically for BST's joystick (live-drive) control
invertDirection	Used to invert the logical direction of the slider
Travel Limits	
LOGICAL_HOME_MM=10	Final, calibrated home position in millimetres that the firmware assigns after completing the homing routine. This value becomes the origin for all future motion commands.
SOFT_MIN_MM = 10	The lowest allowed logical position (in millimetres) that the slider is permitted to move to under command prevents fast runs, overshooting and hitting the end of the rail.
SOFT_MAX_MM=345	SOFT_MAX_MM is the highest allowed logical position (in millimetres) that the slider is permitted to reach. This is how you tell the slider how far it can travel and stay out of trouble! Allow a little cushioning here.

Steps_per_rev

This parameter tells the system how many stepper motor steps there are in one full revolution of the motor and is likely to be one of only two possible values depending on your motor specification.

- For stepper motors with 1.8deg of revolution/Step this value should be set to **200**
- For stepper motors with 0.9deg of revolution/Step this value should be set to **400**

Microsteps

This parameter is currently fixed at 64K microsteps / step and cannot currently be changed.

mm_per_rev

This value is pulley/belt dependent. This parameter tells the system how far the carriage moves per one full revolution of the motor. Maths will get you close but will require fine tuning against what you actually measure on the device.

Example: If we are using a GT2 timing pulley and belt where the pulley has 20Teeth.

Formula:

$$\text{Travel (mm)} = \text{Number of Pulley Teeth} \times \text{Belt Pitch}$$

The belt pitch for a GT2 timing belt (distance between the teeth) =2mm

So we get:

$$20 \times 2 = 40\text{mm}$$

However belt tension and mechanical inaccuracies will have an effect on this so be prepared to test and measure the actual distance traveled matches the commanded distance and adjust this value to correct what you are actually measuring on the device. Typically this may be anywhere between 35 and 40mm on your system.

float steps_per_mm

This is the number of encoder counts required to see 1mm of belt movement on the motor.

This then allows us to calculate exactly where the slider is from the encoder counts.

It is the conversion factor between real-world millimetres and the internal floating-point “logical position” provided by the encoder. $\text{logical_mm} = \text{encoder_counts} \times \text{float_per_mm}$ Think of it as a scale factor for the encoder checked against real world moves.

Note: Currently the encoder is only used to check for drift and is not being used for position output. For that we are using step counts only.

Example:

A 14-bit encoder has 16,384 counts per revolution

This can only be set by reading out the encoder values from the firmware when the slider is commanded a measurable move say 200mm and comparing this with the encoder values in order to establish a real world scale factor.

As a starting point:

In the last example we know our belt should move approx 40mm for each revolution of the motor so this should equal 16,384 encoder counts so $16,384/40 = 409.6$

However mechanical inaccuracies, belt tension etc mean the maths will likely not perfectly match the real world measured values so this parameter allows us to tune the encoder read to match the real world measured values on the device. So after testing on my device this value came out as 321.3

maxSpeed_mm_s

This setting limits the maximum speed the slider can be commanded to move in mm/second during absolute position moves. This protects the device from being over driven by the command software. If a stepper motor is over driven it will stall and you will lose position and step count. This value can be adjusted until you see failures then leave a margin for safety..

accel_mm_s2

This value sets the maximum acceleration which can be applied to the slider. As with maxSpeed_mm_s trying to accelerate too fast can cause the motor to stall particularly with heavy payloads. Again you can adjust this value until the motor stalls and the system loses step count.

bst_max_vel_mm_s

This defines the top speed (in mm/s) that BST is allowed to command during joystick or live-drive control. It acts as a safety and feel limit so BST can't accidentally request speeds that are too fast for the mechanics. It differs from maxSpeed_mm_s in that it is specifically for joystick commands.

bst_accel_mm_s2

The acceleration limit used specifically for BST's joystick (live-drive) control limits the maximum acceleration which can be commanded by the joystick. As before, test it with the joystick hard over until the motor fails. This will also depend on the mass of the load so you will have to make allowance for this.

Logical Home offset (mm)

This provides a cushion of space between home and the limit switches.

When the slider homes at startup the carriage runs down the track until it meets and triggers the limit switch. It then backs off the switch a little then approaches again more slowly. This gives us the basis for a more accurate and repeatable home position. Once the system finds the switch on its slow pass it then backs off the switch to a safer distance= **Logical Home offset (mm)** and registers this as home. **Logical Home offset (mm)** is the distance in mm away from the limit switch the system uses as Home position. This gives us a cushion for fast overshoots, preventing the carriage from hitting the end stops during fast runs where it may not be able to stop fast enough. If the carriage does hit an endstop you will likely lose step count and position accuracy. Test this with your intended Max load.

SOFT_MIN_MM

The lowest allowed logical position (in millimetres) that the slider is permitted to move to under command prevents fast runs, overshooting and hitting the end of the rail. This is an additional soft limit at the homing end of the slider.

SOFT_Max_MM

This is effectively the distance the carriage is allowed to move away from home position. This limit should be set to the maximum length of your rail minus a cushion to allow for fast moves. Note this is measured from your home position. Adjust and test under all conditions.

invertDirection

Used to invert the logical direction of the slider